

tf.compat.v1.distribute.ReplicaContext

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/distribute/ReplicaContext

A class with a collection of APIs that can be called in a replica context.

Function Signatures

```
tf.compat.v1.distribute.ReplicaContext( strategy,  
replica_id_in_sync_group )  
strategy = tf.distribute.MirroredStrategy(['GPU:0', 'GPU:1'])
```

Code Examples

```
tf.compat.v1.distribute.ReplicaContext(  
strategy, replica_id_in_sync_group  
)
```

```
tf.compat.v1.distribute.ReplicaContext(  
strategy, replica_id_in_sync_group  
)
```

```
strategy = tf.distribute.MirroredStrategy(['GPU:0', 'GPU:1'])
def func():
    replica_context = tf.distribute.get_replica_context()
    return replica_context.replica_id_in_sync_group
strategy.run(func)
PerReplica:{
0: ,
1:
}
```

```
strategy = tf.distribute.MirroredStrategy(['GPU:0', 'GPU:1'])
```

```
def func():
```

```
    replica_context = tf.distribute.get_replica_context()
```

```
    return replica_context.replica_id_in_sync_group
```

```
strategy.run(func)
```

Documentation

A class with a collection of APIs that can be called in a replica context.

You can use `tf.distribute.get_replica_context` to get an instance of `ReplicaContext`, which can only be called inside the function passed to `tf.distribute.Strategy.run`.

Attributes

This identifies the replica among all replicas that are kept in sync. The value of the replica id can range from 0 to `tf.distribute.ReplicaContext.num_replicas_in_sync - 1`.

Methods

`## all_reduce`

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All-reduces value across all replicas.

It supports batched operations. You can pass a list of values and it attempts to batch them when possible. You can also specify options to indicate the desired batching behavior, e.g. batch the values into multiple packs so that they can better overlap with computations.

Note that all replicas need to participate in the all-reduce, otherwise this operation hangs. Note that if there're multiple all-reduces, they need to execute in the same order on all replicas. Dispatching all-reduce based on conditions is usually error-prone.

Known limitation: if value contains `tf.IndexedSlices`, attempting to compute gradient w.r.t value would result in an error.

This API currently can only be called in the replica context. Other variants to reduce values across replicas are:

- `tf.distribute.StrategyExtended.reduce_to`: the reduce and all-reduce API

in the cross-replica context.

- `tf.distribute.StrategyExtended.batch_reduce_to`: the batched reduce and

all-reduce API in the cross-replica context.

- `tf.distribute.Strategy.reduce`: a more convenient method to reduce

to the host in cross-replica context.

merge_call

[View source](#)

Merge args across replicas and run `merge_fn` in a cross-replica context.

This allows communication and coordination when there are multiple calls to the `step_fn` triggered by a call to `strategy.run(step_fn, ...)`.

See `tf.distribute.Strategy.run` for an explanation.

If not inside a distributed scope, this is equivalent to: